

BiasConsist: Consistency-Guided Bias Tuning for Deepfake Detection

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Abstract. Deepfake detection has achieved remarkable progress, yet generalization to unseen manipulation methods and datasets remains a major challenge. Although vision foundation models such as CLIP provide strong transferable representations, fully fine-tuning them often leads to overfitting and loss of pretrained knowledge. We propose **BiasConsist**, a parameter-efficient framework that combines bias-only adaptation of the CLIP visual encoder with weak-to-strong consistency regularization. By updating only the encoder biases and a lightweight classification head, BiasConsist limits changes to the pretrained backbone, while the consistency objective encourages consistent predictions across differently augmented views of the same image, reducing sensitivity to visual perturbations. Extensive experiments under cross-dataset and cross-method evaluation demonstrate that BiasConsist achieves competitive generalization while requiring only a small fraction of the trainable parameters of full fine-tuning. These results highlight the effectiveness of stable and parameter-efficient adaptation for robust deepfake detection.

Keywords: Deepfake detection, CLIP, parameter-efficient fine-tuning, bias tuning, transformer, cross-domain generalization

1 Introduction

Recent advances in GANs [1] and Diffusion Models [2] have enabled the generation of highly realistic synthetic faces, posing increasing challenges to media authenticity and motivating the development of generalizable deepfake detectors. Traditional detectors, often reliant on CNNs or frequency analysis, are struggling to keep pace as modern generative models eliminate low-level artifacts. This shift has led the field toward vision foundation models such as CLIP [3] and DINO [4]. These models offer robust, global representations that have already shown significant promise in universal detection tasks [5, 6].

However, fully fine-tuning these large models often leads to catastrophic forgetting, which reduces their generalization ability [7, 8, 9, 10]. To address this issue, Parameter-Efficient Fine-Tuning (PEFT) has emerged as a promising alternative. Recent methods such as Effort [5] and GenD [11] achieve strong cross-dataset performance by adapting models in different spaces, such as weight space or feature space. Despite their effectiveness, these approaches introduce additional complexity, raising the question of whether such designs are truly necessary for robust deepfake detection.

We propose BiasConsist, a parameter-efficient framework that combines bias-only adaptation of a pretrained CLIP visual encoder with weak-to-strong consistency regularization for generalizable deepfake detection. By updating only the encoder biases and a lightweight classification head, BiasConsist limits changes to the pretrained backbone without introducing additional adapter modules or requiring subspace decomposition [5]. The consistency objective further encourages stable predictions across weakly and strongly augmented views of the same image. Building on BitFit [7], our contribution lies in coupling constrained CLIP adaptation with prediction-level consistency. Unlike GenD [11], which tunes LayerNorm parameters and employs feature-space regularization, BiasConsist regularizes predictions across augmented views while

restricting backbone updates to bias parameters. We train using source-domain data and evaluate on five unseen datasets and seven unseen manipulation methods without target-domain validation for model selection.

In summary, our key contributions are as follows:

- **BiasConsist framework:** We integrate bias-only CLIP adaptation with weak-to-strong consistency regularization for generalizable deepfake detection.
- **Consistency-guided adaptation:** We show that consistency learning improves cross-method generalization while maintaining comparable cross-dataset performance.
- **Extensive generalization evaluation:** We evaluate BiasConsist on five unseen datasets and seven unseen manipulation methods, achieving competitive generalization with only 0.27M trainable parameters and without target-domain validation for checkpoint selection.

2 Related Work

Fake image detection has become an increasingly urgent problem in modern society. Early studies mainly addressed this task using vision-specific neural networks, such as CNNs, to capture spatial artifacts [12, 13, 14, 15]. Other works explored frequency-domain analysis to identify abnormal traces introduced during the image generation process [12, 13, 16], or relied on face reconstruction, where the input image is compared with its reconstructed counterpart to determine whether it has been manipulated [14].

Recent studies have turned to pretrained vision-language models such as CLIP [3], which is trained on large-scale image-text pairs and therefore encodes rich semantic knowledge. This property makes CLIP an attractive backbone for fake image detection, and many methods adapt it by fine-tuning on real and fake images [6, 17]. Nevertheless, directly fine-tuning CLIP can be costly and may distort the semantic representations learned during pretraining [9]. In addition, such methods usually achieve strong results on in-domain data but degrade noticeably under out-of-domain settings, indicating overfitting and weak generalization. One major reason is that fake images generated by different models often contain different artifacts. As modern generative models continue to improve in realism and diversity, learning a detector that generalizes across unseen generation methods remains highly challenging.

A key challenge in this line of research is how to adapt pretrained models to fake image detection while preserving the general knowledge acquired during pretraining. When a pretrained model is directly fine-tuned on a specific downstream dataset, its original semantic representation may be distorted, reducing the value of pretraining and limiting its ability to generalize. To address this issue, recent studies have explored parameter-efficient adaptation strategies for vision-language models such as CLIP. For instance, ForAda [18] introduces a lightweight parallel adapter. By freezing the backbone and training only the adapter modules, this method preserves the contextual knowledge of the pretrained model while learning forgery-related features and also reducing computational cost.

Alternatively, prompt learning has emerged as a popular way to adapt CLIP, bypassing full-model tuning by keeping the backbone frozen and optimizing only a small set of learnable prompt vectors. This paradigm is exemplified by general-purpose methods like CoOp [19], CoCoOp [20], and MaPLe [21], and has recently been extended to image forensics via frameworks like CLIP-IFDL [22] and IAPL [23] to boost generalization in forgery detection. However, because prompt-based techniques primarily operate by reshaping the input space or adjusting textual conditioning, they do not directly adapt the internal parameters of the visual encoder. Our work departs from this line of thinking by exploring whether a simple bias-only adjustment restricted to the visual encoder is sufficient for robust deepfake detection.

Other parameter-efficient methods restrict adaptation to minimal subspaces to preserve pretrained representations. For instance, Effort [5] applies singular value decomposition (SVD) to freeze high-rank weight components containing pretrained knowledge, tuning only the low-rank parts for forgery cues. Similarly, GenD [11] avoids architectural modifications and updates only the LayerNorm parameters to shift feature distributions. To improve transferability, GenD [11] introduces feature alignment and uniformity regularization [24] to align related samples and prevent representation collapse. Both methods show that localized parameter updates effectively adapt the model while maintaining out-of-domain robustness.

Beyond these parameter-efficient strategies, bias-only tuning offers an even simpler alternative. As demonstrated by BitFit [7], updating solely the bias terms of a pretrained network can yield competitive performance while leaving the vast majority of parameters untouched. Unlike adapter-, prompt-, or decomposition-based methods, this approach bypasses the need for auxiliary modules, prompt tokens, or weight decompositions, making it a lightweight and non-intrusive strategy for preserving pretrained features. Expanding on this concept, we investigate whether restricting adaptation to the bias terms within the CLIP visual encoder is sufficient to achieve strong generalization in deepfake detection. To complement this lightweight adaptation, we incorporate consistency regularization to further improve robustness. While paradigms like UDA [25] and FixMatch [26] have established the effectiveness of weak-to-strong consistency learning in semi-supervised settings, its utility remains largely underexplored in parameter-efficient image forensics. By encouraging stable predictions across differently augmented views of the same image, consistency regularization reduces the detector's reliance on transient, augmentation-specific artifacts and promotes more domain-invariant forgery cues. Consequently, our proposed framework, BiasConsist, couples bias-only CLIP adaptation with weak-to-strong consistency learning, preserving generalizable representations while improving robustness against unseen datasets and manipulation methods.

3 Method

We propose BiasConsist, a simple parameter-efficient fine-tuning framework for robust and generalizable deepfake detection. As illustrated in Figure 1, BiasConsist processes weakly and strongly augmented views using a shared CLIP ViT-L/14 visual encoder [3], based on the Vision Transformer architecture [27] and a shared linear classification head. Following the bias-only adaptation principle of BitFit [7], all non-bias backbone parameters are frozen, while only the encoder biases and classification head are optimized. Inspired by weak-to-strong consistency learning in UDA [25] and FixMatch [26], the detached weak-view prediction guides the strong view through a KL-divergence loss, while both views are supervised by the ground-truth label. This design aims to preserve CLIP’s transferable representation, improve prediction robustness under appearance perturbations, and enhance generalization to unseen datasets and manipulation methods.

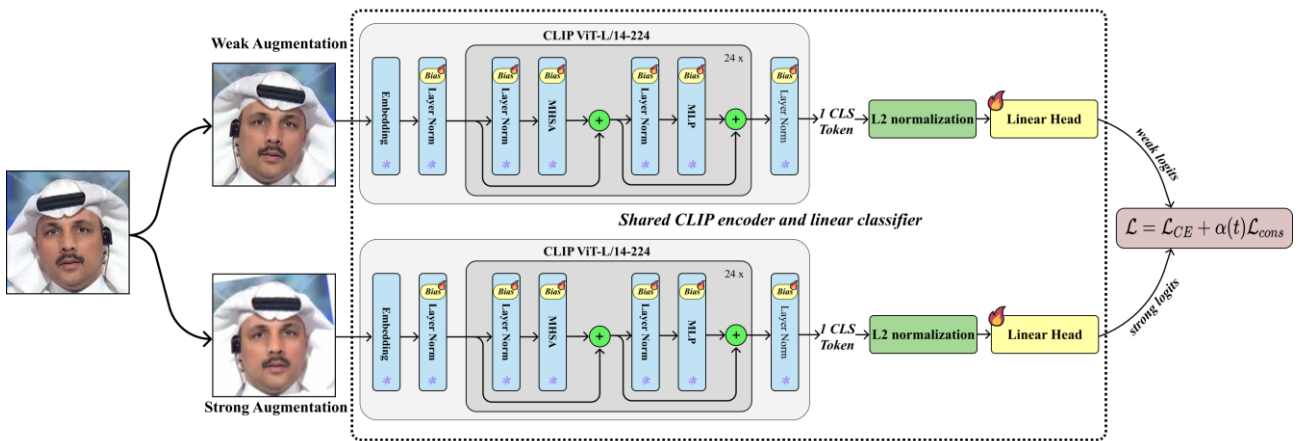


Figure 1. Overview of BiasConsist with bias-only CLIP adaptation and weak-to-strong consistency regularization.

3.1 Feature Extractor

Given an input image x , the CLIP ViT-L/14 visual encoder extracts a global image representation $f = g_\theta(x)$, $f \in \mathbb{R}^{1024}$, where f is the post-LayerNorm CLS token representation produced by the vision encoder. The extracted feature is subsequently L2-normalized before being fed into a lightweight linear classification head for binary deepfake classification.

To preserve the rich visual representations learned during large-scale pretraining and reduce catastrophic forgetting, the CLIP backbone is adapted using a BitFit strategy. Specifically, all backbone weight parameters, including the self-attention weights, feed-forward network weights, and LayerNorm weights, remain frozen throughout training, while only the bias parameters are updated. The final binary prediction is produced by a linear classifier that maps the normalized 1024-dimensional feature vector to two output logits corresponding to the real and fake classes.

3.2 Bias-Only Parameter-Efficient Tuning

CLIP is a powerful pretrained vision-language model with rich contextual representations. However, fully fine-tuning the CLIP backbone is computationally expensive and may lead to catastrophic forgetting. To address this issue, we adopt Bias-Only Parameter-Efficient Tuning (BitFit), which updates only the bias parameters while keeping all backbone weight parameters frozen.

Let the backbone parameters be denoted as $\Theta = \{W, B\}$, where W and B represent the weight and bias parameters, respectively. The optimization process is defined as

$$W^* = W, \quad B^* = B - \eta \frac{\partial \mathcal{L}}{\partial B} \quad (1)$$

where η is the learning rate and $\mathcal{L}$ is the training loss. In our implementation, BitFit is applied to the CLIP ViT-L/14 backbone by freezing all weight parameters of the self-attention, feed-forward, and LayerNorm layers, while updating only their bias parameters. The final linear classification head is fully trainable to adapt the model to the binary deepfake detection task.

3.3 Weak-to-Strong Consistency Regularization

To improve robustness and alleviate overfitting, we employ Weak-to-Strong Consistency Regularization. Specifically, each input image x is transformed into a weak view and a strong view using the augmentation functions $\mathcal{A}_w$ and $\mathcal{A}_s$, respectively. The weak augmentation only performs horizontal flipping with probability $p = 0.5$, whereas the strong augmentation additionally includes random rotation, blur, brightness and contrast adjustments, and image compression. Both views are then fed into the shared model f_θ , yielding $x^w = \mathcal{A}_w(x)$, $x^s = \mathcal{A}_s(x)$, $z^w = f_\theta(x^w)$, $z^s = f_\theta(x^s)$, where the prediction from the weak view is treated as the teacher distribution, while the prediction from the strong view serves as the student distribution.

The teacher distribution is obtained by applying the softmax function with temperature scaling: $p^w = \text{Softmax}\left(\frac{z^w}{\tau}\right)$ and $p^s = \text{Softmax}\left(\frac{z^s}{\tau}\right)$, where τ denotes the temperature parameter. During training, the teacher distribution is detached from the computational graph, so gradients are propagated only through the student branch. This encourages the model to produce consistent predictions for different augmented views of the same input while preventing unstable optimization caused by simultaneously updating both branches.

The consistency loss is defined as the Kullback-Leibler (KL) divergence between the teacher and student distributions:

$$\mathcal{L}_{cons} = \tau^2 D_{KL}(p^w \parallel p^s) \quad (2)$$

During training, the cross-entropy loss is computed for both the weak and strong views:

$$\mathcal{L}_{CE} = \frac{1}{2}(\mathcal{L}_{CE}(z^w, y) + \mathcal{L}_{CE}(z^s, y)) \quad (3)$$

During inference, the model processes only a single input image, and the consistency loss is no longer applied. As a result, the inference cost remains comparable to that of a standard image classification model. The overall training objective combines the supervised cross-entropy loss with the consistency loss:

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha(t)\mathcal{L}_{cons} \quad (4)$$

where $\alpha(t)$ is the consistency weight at epoch t . It is gradually increased using a warm-up schedule to stabilize the early training stage:

$$\alpha(t) = \alpha_{max} \cdot \text{clip}\left(\frac{t - t_{start}}{T_{warmup}}, 0, 1\right) \quad (5)$$

In the configuration used in this study, $\alpha_{max} = 0.5$, $t_{start} = 0$, and $T_{warmup} = 3$.

While one might argue that heavy degradations such as Gaussian blur and JPEG compression in the strong augmentation branch could destroy subtle high-frequency forensic cues, we posit that this is a critical simulation of real-world scenarios. Deepfake videos disseminated across social media platforms inevitably undergo severe compression and resolution degradation. By enforcing prediction consistency across these views, the model is encouraged to reduce its dependence on fragile low-level artifacts and maintain more stable predictions under visual degradation.

3.4 Training and Data Setup

Our proposed model **BiasConsist** is trained on the FaceForensics++ (FF++) dataset under the c23 compression setting and evaluated under cross-dataset and cross-method protocols.

Data Preprocessing

We adopt the standard data preprocessing pipeline from DeepfakeBench [28], executing the workflow as follows:

- Sample 32 frames evenly from each video.
- Extract the largest face in each frame using RetinaFace [29].
- Align the detected face using the predicted facial landmarks.
- Compute the bounding box for the aligned face.
- Enlarge the bounding box by a factor of $1.3\times$.
- Crop the face region and resize it to 224×224 pixels.
- Save the processed image in lossless format.

Weak-to-Strong Data Augmentation

During training, we generate two augmented views from each preprocessed face image. The weak view applies only random horizontal flipping with a probability of 0.5. For the strong view, we follow the augmentation settings in Effort [5] and GenD [11]. This view includes horizontal flipping, rotation from -10° to $+10^\circ$, Gaussian blur with a kernel size of 3-7, random color perturbation, and JPEG compression with quality ranging from 40 to 100. Each augmentation group is applied with a probability of 0.5. These transformations provide noticeable appearance variation while preserving the overall facial structure and are used to construct the strong view for consistency learning.

Training Configuration and Optimization

BiasConsist is trained for 10 epochs with a batch size of 64 using eight data-loading workers. We employ the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, an initial learning rate of 1×10^{-4} , and a weight decay of 1×10^{-4} . The learning rate follows a cosine annealing schedule with $T_{max} = 10$ and a

minimum learning rate of 1×10^{-6} . We use cross-entropy loss with a label-smoothing factor of 0.1 and equal class weights for real and fake samples. The consistency-loss coefficient is linearly increased from 0 to 0.5 over the first three epochs. For the consistency objective, the temperature is set to $\tau = 1.0$, and the KL divergence is computed over all training samples. All experiments are conducted on a single NVIDIA A100 GPU with 80 GB of VRAM.

4 Experiments

This section evaluates the effectiveness and robustness of the proposed BiasConsist framework. Through extensive experiments, we systematically assess the model's in-domain performance, cross-domain generalization, and cross-method transferability, while also validating the underlying architectural design choices through feature-space analyses and ablation studies.

4.1 Datasets

Our model is trained on FaceForensics++ (FF++) C23 [30] and evaluated under three complementary protocols: in-domain, cross-dataset, and cross-method. Table I summarizes the datasets used in this work, together with their roles, descriptions, and the number of forgery generation methods (NoGenMethod). For cross-dataset evaluation, the trained model is directly tested on five unseen benchmarks without using any held-out target-domain validation set for checkpoint selection. For cross-method evaluation, we follow the DF40 [31] benchmark and evaluate seven unseen face-swapping methods, namely UniFace, BlendFace, E4S, FaceDancer, FSGAN, InSwap, and SimSwap, to assess the model's ability to generalize to recently developed manipulation techniques beyond those observed during training.

Table I

DATASETS USED FOR TRAINING AND EVALUATION. NOGENMETHOD DENOTES THE NUMBER OF FORGERY METHODS IN EACH DATASET.

Dataset	Year	Role	NoGenMethod	Description
FF++ [30]	2019	In-domain	4	Main training set with four classical forgery types.
UADFV [32]	2019	Cross-dataset	1	Small benchmark with limited forgery diversity.
CDF-v2 [33]	2020	Cross-dataset	1	High-quality forgeries with subtle real-fake differences.
DFD [34]	2019	Cross-dataset	5	Diverse capture conditions and environments.
DFDCP [35]	2019	Cross-dataset	2	Preview set with quality variation and extra noise.
DFDC [36]	2019	Cross-dataset	8	Large-scale benchmark with compression and real-world variability.
DF40 [31]	2024	Cross-method	40	Broad benchmark for cross-method evaluation.

4.2 Evaluation Benchmark

To ensure a fair comparison with prior state-of-the-art methods, we adopt the Area Under the ROC Curve (AUC) as the primary evaluation metric. Following common approaches in recent deepfake detection benchmarks, we report video-level AUC. The prediction score for each video is obtained by averaging the scores of 32 sampled frames.

4.3 Results

Tables II-IV support the main claim of this paper: BiasConsist achieves strong in-domain performance while maintaining competitive out-of-domain generalization. Rather than aggressively adapting the full backbone, the proposed method preserves the pretrained CLIP representation through bias-only tuning and improves prediction stability through weak-to-strong consistency regularization.

To ensure a fair comparison and avoid redundant computation, we use the in-domain baseline results for ForAda [18], Effort [5], and GenD [11] reported in the original GenD paper [11]. These results are presented

in Table II. We strictly adhere to their experimental protocols, including dataset splits, preprocessing procedures, and evaluation metrics, ensuring the validity and consistency of this comparison.

Table II

IN-DOMAIN EVALUATION OF STATE-OF-THE-ART DETECTORS ON THE FOUR MANIPULATION SUBSETS OF FF++.

Method	DF	F2F	FS	NT	Mean
ForAda [18]	99.7	97.0	98.6	91.9	96.8
Effort [5]	99.4	93.2	98.4	84.6	93.9
GenD (PE) [11]	99.7	99.3	99.1	97.5	98.9
BiasConsist (Ours)	99.9	99.7	98.7	98.2	99.1

In the in-domain FF++ evaluation reported in Table II, the proposed method achieves a mean video-level AUC of 0.991, outperforming all compared parameter-efficient approaches and achieving the best overall performance. In particular, BiasConsist obtains the highest AUC on the DeepFake (99.9%), Face2Face (99.7%), and NeuralTextures (98.2%) subsets, while remaining highly competitive on FaceSwap (98.7%). These results indicate that updating only the bias parameters of the CLIP backbone is sufficient to effectively adapt the pretrained representation to the deepfake detection task without sacrificing discriminative capability.

Table III

CROSS-DATASET EVALUATION RESULTS. ALL METHODS ARE TRAINED ON FF++ C23 [30].

RESULTS ARE OBTAINED FROM THE AUTHORS’ RELEASED CHECKPOINTS OR REPRODUCED USING THEIR OFFICIAL IMPLEMENTATIONS.

Method	Trainable Params	Cross-dataset					
		CDF-v2	DFD	DFDC	DFDCP	UADFV	Avg(D)
F3Net [12]	22M	0.789	0.844	0.718	0.749	0.934	0.807
SPSL [13]	21M	0.799	0.871	0.724	0.770	0.942	0.821
SRM [16]	55M	0.840	0.885	0.695	0.728	0.942	0.818
CORE [37]	22M	0.809	0.882	0.721	0.720	0.941	0.815
RECCE [14]	48M	0.823	0.891	0.696	0.734	0.945	0.818
SLADD [38]	21M	0.837	0.904	0.772	0.756	–	0.817
SBI [15]	18M	0.886	0.827	0.717	0.848	–	0.820
UCF [17]	47M	0.837	0.867	0.742	0.770	–	0.804
IID [39]	66M	0.838	0.939	0.700	0.689	–	0.792
LSDA [40]	133M	0.875	0.881	0.701	0.812	–	0.817
ProDet [41]	96M	0.926	0.901	0.707	0.828	–	0.841
CDFA [42]	87M	0.938	0.954	0.830	0.881	–	0.901
Effort [5]	0.19M	0.956	0.965	0.843	0.909	0.974	0.929
ForAda [18]	5.7M	0.957	0.972	0.872	0.928	0.994	0.945
GenD [11]	0.1M	0.960	0.970	0.870	0.917	0.992	0.942
BiasConsist (Ours)	0.27M	0.966	0.979	0.847	0.868	0.985	0.929

Table III evaluates the cross-dataset generalization of BiasConsist across five unseen benchmarks. The proposed method achieves competitive performance overall, obtaining AUC scores of 0.966 on CDF-v2 and 0.979 on DFD, with an average AUC of 0.929 across the five datasets. These results are achieved while updating only 0.27M trainable parameters. Overall, the findings indicate that bias-only adaptation can preserve transferable CLIP representations under dataset-level domain shifts.

Table IV

CROSS-METHOD EVALUATION RESULTS ON SEVEN UNSEEN MANIPULATION METHODS.

ALL METHODS ARE TRAINED ON FF++ C23 [30].

Method	Trainable Params	Cross-method							
		UniFace	BlendFace	E4S	FaceDan	FSGAN	InSwap	SimSwap	Avg(M)
F3Net [12]	22M	0.809	0.808	0.494	0.717	0.845	0.757	0.674	0.729
SPSL [13]	21M	0.747	0.748	0.514	0.666	0.812	0.643	0.665	0.685
SRM [16]	55M	0.749	0.704	0.704	0.659	0.772	0.793	0.694	0.725
CORE [37]	22M	0.871	0.843	0.679	0.774	0.958	0.855	0.724	0.815
RECCE [14]	48M	0.898	0.832	0.683	0.848	0.949	0.848	0.768	0.832
SLADD [38]	21M	0.878	0.882	0.765	0.825	0.943	0.879	0.794	0.852
SBI [15]	18M	0.724	0.891	0.750	0.594	0.803	0.712	0.701	0.739
UCF [17]	47M	0.831	0.827	0.731	0.862	0.937	0.809	0.647	0.806
IID [39]	66M	0.839	0.789	0.766	0.844	0.927	0.789	0.644	0.800
LSDA [40]	133M	0.872	0.875	0.694	0.721	0.939	0.855	0.793	0.821
ProDet [41]	96M	0.908	0.929	0.771	0.747	0.928	0.837	0.844	0.852
CDFA [42]	87M	0.762	0.756	0.631	0.803	0.942	0.772	0.757	0.775
Effort [5]	0.19M	0.962	0.873	0.983	0.926	0.957	0.936	0.926	0.938
ForAda [18]	5.7M	0.950	0.880	0.968	0.945	0.975	0.948	0.912	0.940
GenD [11]	0.1M	0.964	0.912	0.988	0.954	0.974	0.965	0.944	0.957
BiasConsist (Ours)	0.27M	0.977	0.930	0.990	0.972	0.978	0.969	0.960	0.968

Table IV further evaluates the proposed method under the more challenging cross-method setting. BiasConsist achieves the strongest generalization to unseen manipulation methods among the compared parameter-efficient approaches, with an average AUC of 0.968, compared with 0.957 for GenD, 0.940 for ForAda, and 0.938 for Effort. It also achieves the highest AUC among these methods across all seven unseen manipulation techniques. These results suggest that weak-to-strong consistency regularization helps improve robustness to previously unseen forgery methods by encouraging more stable predictions under visual perturbations. Moreover, these results are achieved while updating only 0.27M trainable parameters, demonstrating that effective adaptation can be obtained without extensive modification of the pretrained CLIP backbone.

Overall, the experimental results demonstrate that robust deepfake detection does not necessarily require full fine-tuning or sophisticated adaptation modules. By combining lightweight bias-only adaptation with weak-to-strong consistency regularization, the proposed framework effectively preserves the transferable representations of CLIP while maintaining competitive cross-dataset performance and improving generalization to unseen manipulation methods.

4.4 Feature-Space Comparison on Celeb-DF-v2

To analyze the learned feature spaces, we feed the Celeb-DF-v2 test samples into each model and extract the embeddings from the final feature layer before the classifier. We then project these high-dimensional embeddings into two dimensions using t-SNE. As shown in Figure 2, Full FT, GenD [11], and Effort [5] exhibit considerable overlap between real and fake samples. In contrast, BiasConsist shows a more compact projected distribution, with real and fake samples tending toward separation, although some overlap remains. These observations provide a qualitative illustration of the learned representations, while the

quantitative cross-dataset results in Table III serve as the primary evidence of generalization performance.

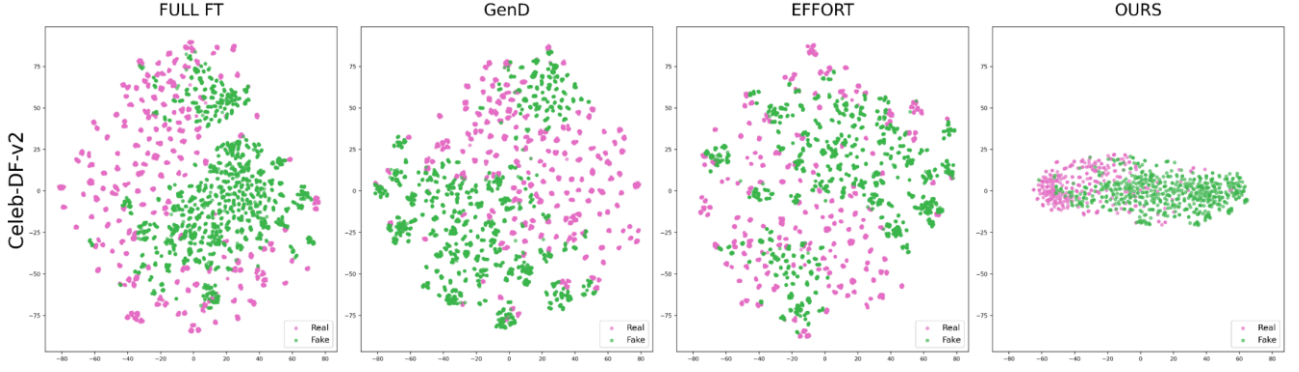


Figure 2. Two-dimensional t-SNE projections of real and fake samples from Celeb-DF-v2 in the feature spaces learned by Full FT, GenD, Effort, and BiasConsist. BiasConsist exhibits a more compact projected distribution, with a visible tendency toward separation between real and fake samples.

4.5 Ablation Study

To evaluate the contribution of each component, we conduct two ablation studies under cross-dataset and cross-method settings. The first compares full fine-tuning with three parameter-efficient fine-tuning (PEFT) strategies and examines the effect of L_2 feature normalization. The second isolates the contribution of the proposed weak-to-strong consistency regularization.

Table V evaluates all configurations on the same five cross-dataset benchmarks used in the main evaluation: CDF-v2, DFD, DFDC, DFDCP, and UADFV. The reported average is computed over all five benchmarks, ensuring consistent benchmark coverage across the ablation configurations.

Table V

ABLATION OF PEFT STRATEGIES AND L_2 FEATURE ON FIVE CROSS-DATASET BENCHMARKS. VIDEO-LEVEL AUC IS REPORTED AS MEAN $\pm$ STANDARD DEVIATION ON FIVE TRAINING SEEDS.

Method	Head	L_2 Norm	Loss	Cross-dataset					Avg
				CDF-v2	DFD	DFDC	DFDCP	UADFV	
Full FT	Linear	No	CE	0.655 $\pm$ 0.032	0.495 $\pm$ 0.028	0.560 $\pm$ 0.041	0.612 $\pm$ 0.030	0.624 $\pm$ 0.036	0.589 $\pm$ 0.033
LoRA	Linear	No	CE	0.941 $\pm$ 0.007	0.977 $\pm$ 0.005	0.844 $\pm$ 0.010	0.855 $\pm$ 0.008	0.988 $\pm$ 0.004	0.921 $\pm$ 0.007
LoRA	Linear	Yes	CE	0.893 $\pm$ 0.009	0.955 $\pm$ 0.008	0.819 $\pm$ 0.012	0.842 $\pm$ 0.010	0.986 $\pm$ 0.005	0.899 $\pm$ 0.009
Bias-only	Linear	No	CE	0.938 $\pm$ 0.018	0.958 $\pm$ 0.016	0.831 $\pm$ 0.022	0.858 $\pm$ 0.014	0.990 $\pm$ 0.012	0.915 $\pm$ 0.016
Bias-only	Linear	Yes	CE	0.956 $\pm$ 0.020	0.978 $\pm$ 0.017	0.859 $\pm$ 0.024	0.877$\pm$0.013	0.988 $\pm$ 0.013	0.932$\pm$0.017
LayerNorm	Linear	No	CE	0.931 $\pm$ 0.010	0.966 $\pm$ 0.009	0.840 $\pm$ 0.012	0.852 $\pm$ 0.009	0.993$\pm$0.006	0.916 $\pm$ 0.009
LayerNorm	Linear	Yes	CE	0.942 $\pm$ 0.010	0.967 $\pm$ 0.009	0.860$\pm$0.012	0.858 $\pm$ 0.009	0.990 $\pm$ 0.007	0.923 $\pm$ 0.009
BiasConsist	Linear	Yes	CE+KL	0.966$\pm$0.008	0.979$\pm$0.006	0.847 $\pm$ 0.011	0.868 $\pm$ 0.015	0.985 $\pm$ 0.007	0.929 $\pm$ 0.009

Table V shows that full fine-tuning exhibits the weakest cross-dataset generalization, whereas the PEFT strategies substantially improve performance. Among them, LoRA and LayerNorm tuning achieve average AUCs nearly 0.90, while bias-only tuning with L_2 feature normalization achieves the highest average AUC of 0.932 ± 0.017 . BiasConsist obtains a comparable 0.929 ± 0.009 . Although weak-to-strong consistency does not further improve the average cross-dataset performance, both bias-based configurations exhibit low standard deviations, indicating stable results across different training seeds. These results suggest that lightweight adaptation is effective for cross-dataset generalization, while broader dataset-level shifts may involve variations that are not directly addressed by the consistency objective alone.

Table VI

EFFECT OF WEAK-TO-STRONG CONSISTENCY ON CROSS-METHOD VIDEO-LEVEL AUC.

RESULTS ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION.

Method	Cross-method							
	UniFace	BlendFace	E4S	FaceDan	FSGAN	InSwap	SimSwap	Avg(M)
Bias-only	0.967 $\pm$ 0.011	0.916 $\pm$ 0.015	0.984 $\pm$ 0.011	0.956 $\pm$ 0.011	0.976 $\pm$ 0.008	0.952 $\pm$ 0.015	0.947 $\pm$ 0.014	0.957 $\pm$ 0.011
BiasConsist	0.977$\pm$0.005	0.930$\pm$0.009	0.990$\pm$0.003	0.972$\pm$0.008	0.978$\pm$0.007	0.969$\pm$0.009	0.960$\pm$0.008	0.968$\pm$0.006

Table VI further evaluates the effect of the proposed weak-to-strong consistency regularization on unseen manipulation methods. Incorporating the consistency objective consistently improves performance across all seven manipulation methods, increasing the average AUC from 0.957 to 0.968. These improvements suggest that prediction consistency reduces sensitivity to method-specific artifacts and promotes cues that transfer more effectively to unseen forgery techniques.

This contrast suggests that consistency regularization is more beneficial when the domain shift is primarily associated with unseen manipulation techniques, whereas broader dataset-level shifts involve additional sources of variation that are not directly addressed by the consistency objective.

5 Conclusion

We present BiasConsist, a simple and parameter-efficient approach to generalizable deepfake detection. Instead of fully fine-tuning a pretrained foundation model or adding complex adapter modules, BiasConsist updates only the biases of the CLIP ViT-L/14 encoder and all parameters of a lightweight classification head, while keeping the remaining encoder parameters frozen. To reduce overfitting to fragile, generator-specific artifacts, we introduce a weak-to-strong consistency objective that encourages stable predictions under visual perturbations. Experiments show that BiasConsist achieves competitive cross-dataset performance with only 0.27M trainable parameters, while ablation results indicate that consistency regularization improves cross-method generalization. These findings support the effectiveness of combining constrained adaptation with prediction consistency for parameter-efficient deepfake detection. Currently, BiasConsist operates on individual frames and relies on handcrafted augmentations. Future work will explore video-level architectures to incorporate temporal cues, adaptive parameter-efficient mechanisms, and pair-aware consistency learning.

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